Reliability-aware Spatial Agents Orchestration

1. Adaptive Role Assignment

Input (image, query)



Which object is higher in 3D world space, the cyclist in orange suit or the yellow board?

Orchestrator

Classify task

"Spatial Relation"

Select top-3 agents and return their role with confidence



Specific Role & Profile

- Implicit Visual Reasoning
- Explicit 3D Reconstruction
- Scene-graph Construction

Avg. Confidence Score

Specialist A	0.90
Specialist B	0.84
Specialist C	0.79
Specialist D	0.65
Specialist E	0.50

Time step: t

Update confidence score after Test-Time Orchestration.

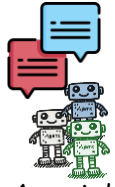
2. Role-conditioned Specialist Execution

	Intermediate representations		Answer
Specialist A	Depth map		The yellow board
Specialist B	Scene-graph		The cyclist
Specialist C	Reasoning traces	The cyclist is in the foreground, currently at the peak of a jump..	The yellow board

3. Reliability-aware Final-Reasoning



Input



Agents' output



Confidence score

Reasoning Agent

Final answer